Chi-Squared (χ²) Analysis

A Comprehensive Technical Overview

Chi-squared (χ^2) analysis is a statistical method used to determine if there is a significant difference between **observed frequencies** and **expected frequencies** in categorical data. It is a non-parametric test used for nominal or ordinal variables.

1. Core Applications

- **Goodness of Fit:** Evaluates if a sample data distribution matches a theoretical population distribution.
- **Test of Independence:** Determines whether two categorical variables are likely to be related or if they are independent.

2. The Mathematical Foundation

The Chi-squared statistic measures the discrepancy between the observed counts and the counts expected under the null hypothesis.

$$\chi^2 = \Sigma \left[(O_i - E_i)^2 / E_i \right]$$

Where:

- O_i = Observed frequency (actual count)
- E_i = Expected frequency (theoretical count)

3. Example 1: Goodness of Fit

Scenario: A bakery claims equal sales across four cookie types. 100 total sales are observed.

Cookie Type	Observed (O)	Expected (E)
Chocolate Chip	35	25
Oatmeal	20	25
Sugar	15	25
Peanut Butter	30	25

Analysis Results:

- **Degrees of Freedom (df):** $n - 1 = 3$
- **χ^2 Calculation:** $4 + 1 + 4 + 1 = 10.0$
- **Critical Value (α = 0.05):** 7.81
- **Conclusion:** Since $10.0 > 7.81$, the null hypothesis is rejected. Sales are not equally distributed.

4. Example 2: Test of Independence

Scenario: Testing if employee work mode (Remote/Office) is independent of department (Engineering/Marketing).

Observed Contingency Table

Mode / Dept	Engineering	Marketing	Total
Remote	60	40	100
In-Office	20	80	100
Total	**80**	**120**	**200**

Analysis Results:

- **Expected Value Formula:** *$E = (\text{Row Total} \times \text{Col Total}) / \text{Grand Total}$*
- **χ^2 Calculation:** *≈ 33.34*
- **Degrees of Freedom:** *$(2-1) \times (2-1) = 1$*
- **Conclusion:** Significant relationship exists ($p < 0.05$). Engineers show a significantly higher frequency of remote work in this sample.

5. Key Assumptions

- **Independence:** Each observation is independent of the others.
- **Sample Size:** Every cell in the expected frequency table should ideally be ≥ 5.
- **Data Type:** Must use raw counts/frequencies, not percentages or means.